

Surname _____

Van _____

Initials _____

Voorletters _____

Stud. No/nr. _____

The undersigned hereby confirms that he/she is familiar with and accepts the examination regulations

Ondergetekende bevestig hiermee dat hy/sy op hoogte is met die eksamenregulasies en dat hy/sy hom/haar daaraan onderwerp

Handtekening _____ Signature _____ Date _____ Datum _____

$v_{avg} = \frac{1}{2}(v_0 + v)$	$v_x^2 = v_{ox}^2 + 2a_x(x - x_0)$	$a = \frac{v^2}{r}$	$R = \frac{v_0^2}{g} \sin 2\theta_0$
$v = v_0 + at$	$x - x_0 = \frac{1}{2}(v_{0x} + v_x)t$	$g = 9,8 \text{ m/s}^2$	$T = \frac{2v_0 \sin \theta}{g}$
$x = x_0 + v_0 t + \frac{1}{2}a_x t^2$	$x - x_0 = v_x t - \frac{1}{2}a_x t^2$	$h = \frac{v_0^2 \sin^2 \theta}{2g}$	
$x = x_0 + v_{avg,x} t$			

Question 1/Vraag 1 ch 2:17 +(d)

The position of a particle moving along the x -axis is given in centimetres by $x = 9.75 + 1.50t^3$, where t is in seconds. Calculate the following:

Die posisie van 'n deeltjie wat langs die x -as beweeg word in sentimeter gegee deur $x = 9.75 + 1.50t^3$, waar t in sekondes is. Bereken die volgende:

- (a) The average velocity during the time interval $t = 2.00 \text{ s}$ to $t = 3.00 \text{ s}$.

Die gemiddelde snelheid gedurende die tydsinterval $t = 2.00 \text{ s}$ tot $t = 3.00 \text{ s}$. [5]

$x = 9.75 + 1.50t^3$
 For $t = 2.00 \text{ s}$ we get $x_2 = 21.75 \text{ cm}$
 And $t = 3.00 \text{ s}$ and $x_3 = 50.25 \text{ cm}$. Using the definition for average velocity for the time interval $2.00 \leq t \leq 3.00 \text{ s}$:

$$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{50.25 \text{ cm} - 21.75 \text{ cm}}{3.00 \text{ s} - 2.00 \text{ s}}$$

which yields $v_{avg} = 28.5 \text{ cm/s}$.

- (b) The instantaneous velocity at $t = 2.00 \text{ s}$.

Die oombliklike snelheid by $t = 2.00 \text{ s}$. [3]

$$v = \frac{dx}{dt} = 4.5t^2$$

At $t = 2.00 \text{ s}$ $v = 4.5t^2 = 4.5(2\text{s})^2 = 18.00 \text{ cm/s}$

- (c) The instantaneous velocity when the particle is midway between its positions at $t = 2.00 \text{ s}$ to $t = 3.00 \text{ s}$.

Die oombliklike snelheid wanneer die deeltjie by die posisie is wat presies halfpad tussen $t = 2.00 \text{ s}$ en $t = 3.00 \text{ s}$ is. [5]

Position midway: $\Delta x = 50.25 - 21.75 = 28.5 \text{ cm}$ and position is therefore $x = 21.75 + \frac{1}{2}(28.5) = 36 \text{ cm}$
 Time when at midway position:

$$x = 9.75 + 1.50t^3$$

$$t = \sqrt[3]{\frac{x - 9.75}{1.50}} = \sqrt[3]{\frac{36.00 - 9.75}{1.50}} = \sqrt[3]{17.5} = 2.60 \text{ s}$$

Instantaneous velocity at this time: $v = 4.5t^2 = 4.5(2.60\text{s})^2 = 30.42 \text{ cm/s}$

(d) The instantaneous acceleration at $t = 2.00 \text{ s}$.

Die oombliklike versnelling by $t = 2.00 \text{ s}$.

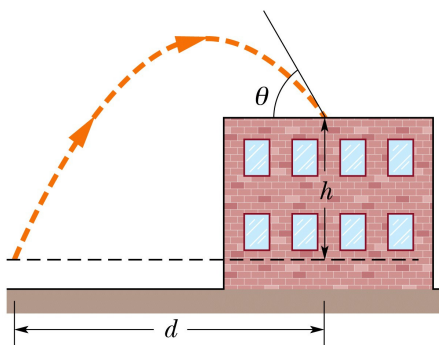
[3]

$$a = \frac{dv}{dt} = (2)4.5t$$

At $t = 2.00 \text{ s}$

$$a = 9(2.00) = 18.00 \text{ cm/s}^2$$

Question 2 / Vraag 2 Ch 4: 48



A ball is thrown up onto a roof, landing 4.50 s later at a height $h = 20 \text{ m}$ above the release level; refer to the figure. The ball's path just before landing is angled at $\theta = 60^\circ$ with the roof.

'n Bal word tot op 'n dak gegooi, en land 4.50 s later op 'n hoogte van $h = 20 \text{ m}$ bo die lanseervla; verwys na die figuur. Die bal se pad net voor dit land maak 'n hoek van $\theta = 60^\circ$ met die dak.

(a) Determine the horizontal distance d the ball travels.

Bepaal die horisontale afstand d wat die bal aflê.

[5]

$t = 4.50 \text{ s}$ $a_y = -9.8 \text{ m/s}^2$ $y = 20 \text{ m}$ $v_{oy} = ?$ $v_y = ? \text{ at } 4.5\text{s}$	$a_x = 0 \text{ m/s}^2$ $x = ?$ $v_{ox} = ?$
$\Delta y = v_y t - \frac{1}{2} a t^2$ $v_y = \frac{\Delta y + \frac{1}{2} a t^2}{t} = \frac{+20 \text{ m} + \frac{1}{2} (-9.8 \text{ m/s}^2) (4.50 \text{ s})^2}{4.50 \text{ s}}$ $= -17.60 \text{ m/s}$ $\tan \theta = \frac{v_y}{v_x}$ $v_x = \frac{v_y}{\tan \theta} = \frac{17.6}{\tan 60}$ $= 10.16 \text{ m/s}$ Horizontal distance: $\Delta x = v_x t - \frac{1}{2} a_x t^2$ $= 10.16 \text{ m/s} (4.5 \text{ s}) = 45.72 \text{ m}$	

- (b) What is the magnitude and direction (relative to the horizontal) of the ball's initial velocity?
 Wat is die grootte en rigting (relatief tot die horisontale as) van die bal se aanvanklike snelheid?

[5]

$$\Delta y = v_{oy}t + \frac{1}{2}at^2$$

$$v_{oy} = \frac{\Delta y - \frac{1}{2}at^2}{t} = \frac{+20 \text{ m} - \frac{1}{2}(-9.8 \text{ m/s}^2)(4.50 \text{ s})^2}{4.50 \text{ s}}$$

$$= 26.50 \text{ m/s}$$

No acceleration in x-direction therefore $v_{ox} = v_x = 10.16 \text{ m/s}$

Magnitude of initial velocity: $v_o = \sqrt{v_{ox}^2 + v_{oy}^2} = 28.4 \text{ m/s}$

Direction:

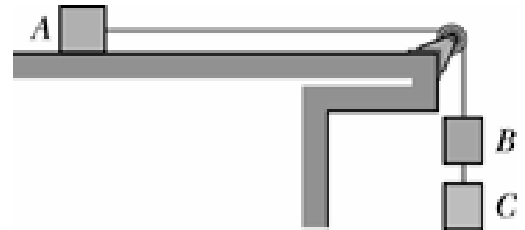
$$\tan \theta = \frac{v_{oy}}{v_{ox}}$$

$$\theta = 68.9^\circ \text{ above the horizontal}$$

Question 3 / Vraag 3 Ch 5: 50 h/w

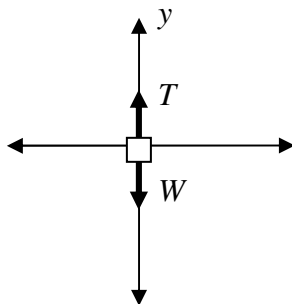
In the Figure, three boxes are connected by cords, one of which wraps over a pulley having negligible friction on its axle and negligible mass. The three masses are $m_A = 30.0 \text{ kg}$, $m_B = 30.0 \text{ kg}$, and $m_C = 10.0 \text{ kg}$. When the assembly is released from rest determine the tension in the cord connecting B and C.

Die Figuur dui drie bokse aan wat verbind word deur toue. Een van die toue loop oor 'n katrol met weglaatbare wrywing op die as en weglaatbare massa. Die massas van die bokse is $m_A = 30.0 \text{ kg}$, $m_B = 30.0 \text{ kg}$, en $m_C = 10.0 \text{ kg}$. Bepaal die spankrag in die tou tussen B en C wanneer die sisteem uit rus losgelaat word.



[2+5]

For box C



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The net force on the system (of total mass $M = 70.0 \text{ kg}$) is the force of gravity acting on the total overhanging mass ($m_{BC} = 40.0 \text{ kg}$).

$$\sum F = Ma = m_{B+C} g$$

$$a = \frac{m_{B+C} g}{M} = \frac{40 \text{ kg}(9.8 \text{ m/s}^2)}{70 \text{ kg}} = 5.6 \text{ m/s}^2$$

This acceleration is directed to the right for box A and downwards for blocks B and C.

Next we apply Newton's second law to box C itself

$$\sum F_y = -m_c a = T - W$$

$$T = -m_c a + m_c g = -(10 \text{ kg})(5.6 \text{ m/s}^2) + (10 \text{ kg})(9.8 \text{ m/s}^2)$$

$$= 42.0 \text{ N}$$